Lecture 11 - Calculus

COMP3670/COMP6670 - Introduction to Machine Learning

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Recall the end of Lecture 4

- Lectures 2 and 3 settled **whether** learning is possible, and Lecture 4 provided the **models**. But almost every one of them ended at the same place: *"and now minimise this"*
- **Maximum likelihood**: for the coin we read $\hat{\theta} = 7/10$ off a table of five candidate values, and promised to learn *"how to compute maximum likelihood estimators"*
- **Logistic regression**: minimise the cross-entropy loss, for which there is *"no closed form"*
- **Least squares**: make the residual sum of squares as small as possible
- The **support vector machine** was left to *"numerical methods"*, with the promise that the basics of **convex optimisation** would come *"later in this course"*

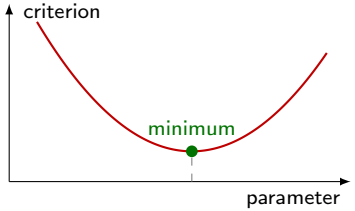
The open question

We know *what* to minimise. How do we actually **compute a minimiser**?

Optimise a function of the parameters

	what is fixed	what varies	what we are after
The coin (Lecture 4)	7 heads in 10 tosses	θ	maximise $\mathcal{L}(\theta) = 120\,\theta^7(1-\theta)^3$
Logistic regression (Lecture 4)	the sample $\mathcal{D}_N$	$w_0, w_1, \ldots, w_d$	minimise $-\frac{1}{N}\,\mathrm{LL}(\mathbf{w})$, the average cross-entropy loss
Least squares (Lecture 4)	the sample $\mathcal{D}_N$	$\boldsymbol{\theta}$	minimise $\sum_{i=1}^{N}\left(Y_i - h(\mathbf{X}_i;\boldsymbol{\theta})\right)^2$
Soft-margin SVM	the sample and C	$\mathbf{w},\ b$	minimise $\frac{1}{2}\|\mathbf{w}\|^2 + C\sum_i \max\{0,\ 1 - y_i(\langle \mathbf{w}, \mathbf{x}_i\rangle + b)\}$

- In every row **the data is fixed and the parameters vary**, so we have a **function** of the parameters
- Maximising is the same job as minimising since the maximiser of $\mathcal{L}$ is the minimiser of $-\mathcal{L}$
- So *training a model* comes down to one task: **find the minimum of a function**



Fitting a model

- In a previous tutorial, **one line** trained a model:

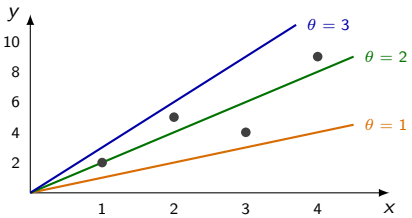
```
model = LinearRegression().fit(features, y_reg)
```

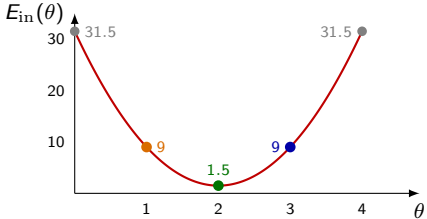
- That line returned the parameters and inside it there was an **optimisation algorithm**
- We have in fact already run three such algorithms by hand:
  - the **PLA**: find a misclassified point, correct the weights, repeat
  - **K-means**: recompute the centroids, reassign the points, repeat
  - the **SVM update**: shrink the weights, correct for the violating points, repeat
- All three have the same structure: **start somewhere, improve a little, repeat**
- In the next lecture we will study general ways of *designing* a step that provides an *improvement* and desirably converges to the minimum of a criterion

These are based on calculus, the subject of this lecture, that provides tools to study the behaviour of functions and characterise their points of minimum and maximum

A running example of linear regression

- One feature, one parameter: a line through the origin, $h_\theta(x) = \theta x, \theta \in \mathbb{R}$, fitted to the four observations $(1,2)$, $(2,5)$, $(3,4)$, $(4,9)$
- Every value of θ is a line, and every line has an in-sample error under the squared loss: $E_{\text{in}}(\theta) = \frac{1}{4}\sum_{i=1}^{4}\left(y_i - \theta x_i\right)^2 = 7.5\,\theta^2 - 30\,\theta + 31.5$





- **Left**: three candidate lines. **Right**: the error of *every* candidate. **Fitting the model means finding the minimum of the curve on the right**
- Here we can simply see it: $\hat{\theta} = 2$. With hundreds of parameters we cannot build a plot and we need algorithms to find the minimum

The plan for this lecture

This lecture: Calculus

The language. Functions and limits, the derivative and its rules, maxima and minima, convexity, several variables

Next lectures: Optimisation

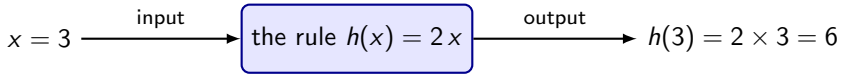
The algorithms. Convex problems, least squares, gradient descent, stochastic gradient descent, Adam

Calculus

Definition of a function

Definition (Function)

A **function** f is a rule that assigns to each **input** x exactly **one output** $f(x)$. We write $f : A \to B$ when the inputs come from the set A and the outputs lie in the set B.



- "Exactly one output" is what makes a rule a function
- Two inputs may share an output, but one input never has two outputs
- A function need not be a formula: a table, a graph or a computer program can be one too

Functions in this course

A hypothesis h

Input: a point x.
Output: a prediction.
The parameter is held *fixed*

$$x \quad \mapsto \quad \theta x$$

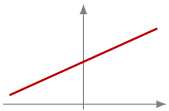
An error E_{in}

Input: the parameter θ.
Output: one number.
The data is held *fixed*

$$\theta \quad \mapsto \quad \frac{1}{4}\sum_{i=1}^{4}(y_i - \theta x_i)^2$$

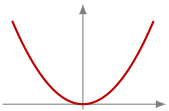
- The *same* expression θx sits inside both, but one is **a function of the data point** x and the other is a **function of the parameter** θ
- The input of the function is clear from its notation, and may vary depending on the context: $f(x), f(t), f(x,t), g(x), h(x), E_{\text{in}}(\theta), L(\theta), L_n(\theta), ...$

Examples of functions



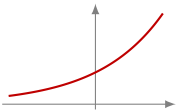
a line

$ax + b$



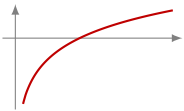
a parabola

x^2



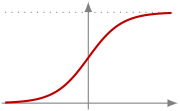
exponential

e^x



logarithm

$\log x$



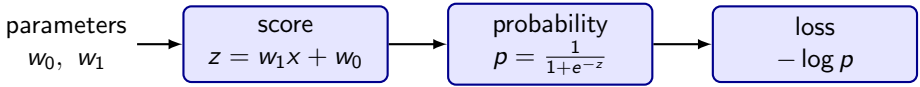
logistic curve

$$\frac{1}{1 + e^{-x}}$$

- log always means the natural logarithm. It **undoes** the exponential: $\log(e^x) = x$
- **Logarithms turn products into sums**: $\log(ab) = \log a + \log b$. That is how the likelihood (a product) became the log-likelihood (a sum) in Lecture 4
- e^x and $\log x$ are both **increasing**: a larger input gives a larger output

Building functions from functions

- Almost every function in this course is **built** from the five on the previous slide, in three ways:
  - **Sum**: $(f+g)(x) = f(x) + g(x)$. An in-sample error is a sum, one term per observation
  - **Product**: $(fg)(x) = f(x)\,g(x)$. The likelihood of the coin, $\theta^7(1-\theta)^3$
  - **Composition**: $(g \circ f)(x) = g\big(f(x)\big)$, "apply f, then g". The squared loss $(y - \theta x)^2$: first the residual $y - \theta x$, then the square



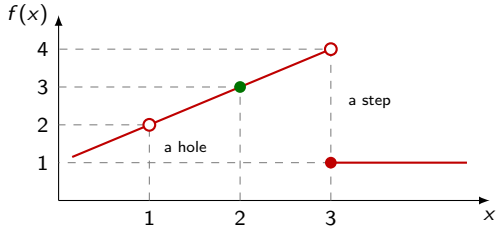
A model as a pipeline of functions

The picture is logistic regression on one observation with label 1: **a line, then the logistic curve, then a logarithm**

What happens near a point: the limit

- Take the function

$$f(x) = \begin{cases} \dfrac{x^2-1}{x-1} & \text{if } x < 3 \\ 1 & \text{if } x \geq 3 \end{cases}$$



- What does $f(x)$ do as x gets **closer and closer** to a point, from either side, without ever being equal to it?

as x approaches	**from the left**	**from the right**	**the limit of $f(x)$**
2	3	3	is 3, which is also the value $f(2)$
1	2	2	is 2, never reached: $f(1)$ is not defined
3	4	1	does not exist: the two sides disagree

- We write $\lim_{x \to 1} f(x) = 2$: **the limit of $f(x)$ as x approaches 1 is 2**

The definition of limit

The limit

We write $\lim_{x \to a} f(x) = L$ when $f(x)$ can be made **as close to L as we like** by taking x **close enough to** a, but different from a.

- The value of f *at* a plays no part. $f(a)$ may be undefined, as on the previous slide, or even different from L. A limit is about **approaching**, never about arriving
- A limit may **fail to exist**: at the step of the previous slide, the two sides approach different numbers
- We have met a limit already. In Lecture 2 the definite integral was the **limit** of Riemann sums as the rectangles became thinner, $\Delta x \to 0$
- The precise definition of limit, with "its ϵ and δ" is out of the scope of this course

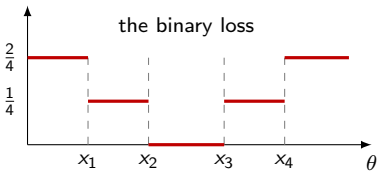
Continuous and discontinuous functions

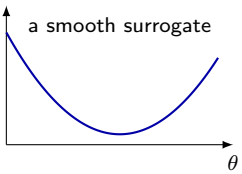
- A function is **continuous** at a if its limit there equals its value, $\lim_{x \to a} f(x) = f(a)$: **a small change of the input produces a small change of the output**. No jumps

- Our five functions are continuous. The **indicator** $\mathbb{1}\{\cdot\}$ is not: it jumps from 0 to 1
- So the in-sample error under the **binary loss** is a **staircase**: each indicator in the formula switches once, at $\theta = x_i$
- In particular, moving the parameter a little in general changes *nothing*

$h_\theta(x) = \mathbb{1}\{x > \theta\}$, points $x_1 < x_2 < x_3 < x_4$, labels 0, 0, 1, 1:

$$\begin{aligned} E_{\text{in}}(\theta) &= \tfrac{1}{4} \textstyle\sum_{i=1}^{4} \mathbb{1}\{h_\theta(x_i) \neq y_i\} \\ &= \tfrac{1}{4}\big[\mathbb{1}\{x_1 > \theta\} + \mathbb{1}\{x_2 > \theta\} + \mathbb{1}\{x_3 \leq \theta\} + \mathbb{1}\{x_4 \leq \theta\}\big] \end{aligned}$$





Hence the **surrogate losses** *of Lecture 4: we need an error that responds smoothly.*

The derivative

How fast is it changing?

Definition (Average rate of change)

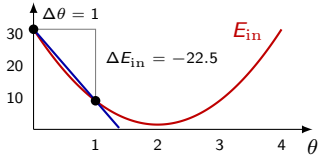
When the input goes from x to $x + \Delta x$, the output changes by $\Delta f = f(x + \Delta x) - f(x)$. The **average rate of change** of f between x and $x + \Delta x$ is the change in the output per unit of change in the input,

$$\frac{\Delta f}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}.$$

It is the slope of the straight line through the two points of the graph, called a **secant**.

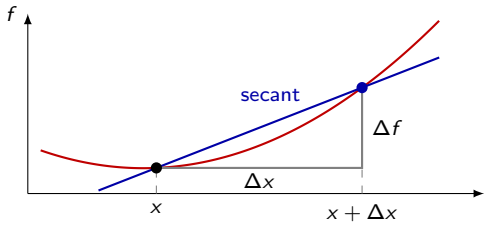
- Running example, $E_{\text{in}}(\theta) = 7.5\,\theta^2 - 30\,\theta + 31.5$:

θ goes from	0 to 1	1 to 2	2 to 3	3 to 4	0 to 4
average rate	-22.5	-7.5	7.5	22.5	0

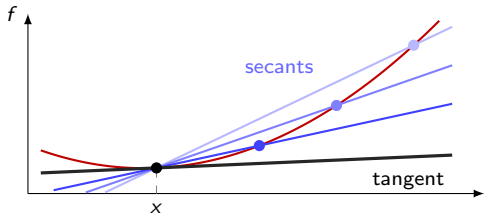


- The rate depends on the interval, and an average hides how the function changes inside it. What is the rate of change **at one value** of θ?

From the secant to the tangent



secant slope $= \Delta f / \Delta x =$ average rate of change



as Δx shrinks, the secants converge to the **tangent**

- **The idea**: keep the point fixed and let the second point slide towards it. The slopes of the secants approach a **limit**: the slope of the tangent
- For $f(x) = x^2$ at $x = 1$ the slopes approach 2. We cannot simply set $\Delta x = 0$: the ratio would read $0/0$

Δx	1	0.1	0.01	0.001
secant slope	3	2.1	2.01	2.001

The derivative

Definition (Derivative)

The **derivative** of f at x is the limit of the average rate of change as the interval shrinks to the point:

$$f'(x) = \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}.$$

When the limit exists, f is said to be **differentiable** at x.

- **In the picture**: $f'(x)$ is the slope of the tangent to the graph at x
- **As a rate**: it is the *instantaneous* rate of change. For an error $E_{\text{in}}(\theta)$, it is how fast the error changes as the parameter moves, at that exact value of θ
- **Notation**: $f'(x)$ is also written $\frac{df}{dx}$, a reminder that it was born as the ratio of a small change in f to a small change in x
- Compare with Lecture 2. The integral was a limit of *sums* of thin rectangles and the derivative is a limit of *ratios* of small changes. Both consider the variation $\Delta x \to 0$

One derivative from the definition

- Let us compute one derivative straight from the definition, for $f(x) = x^2$. The average rate of change is

$$\frac{(x+\Delta x)^2 - x^2}{\Delta x} = \frac{x^2 + 2x\,\Delta x + (\Delta x)^2 - x^2}{\Delta x} = \frac{2x\,\Delta x + (\Delta x)^2}{\Delta x} = 2x + \Delta x$$

- As Δx approaches 0, this approaches $2x$. So **the derivative of x^2 is $2x$**
- Check at $x = 1$: the derivative is 2, the number the secant slopes were approaching. In fact the table two slides ago *is* the formula $2x + \Delta x$ at $x = 1$: 3, 2.1, 2.01, 2.001
- The same computation for the running example, $E_{\text{in}}(\theta) = 7.5\,\theta^2 - 30\,\theta + 31.5$, gives the average rate $15\,\theta - 30 + 7.5\,\Delta\theta$, which approaches $E'_{\text{in}}(\theta) = 15\,\theta - 30$

This is the only limit we will compute in this course. From here on, **rules** *will be provided without proof.*

Three ways to interpret $f'(x)$

A slope

The slope of the tangent to the graph at x

A rate

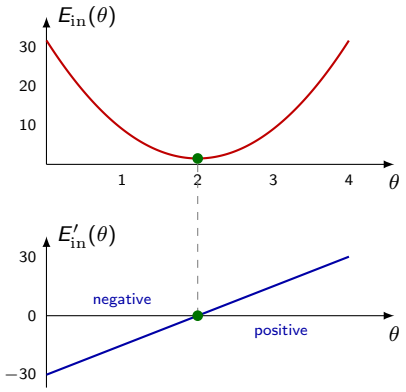
How fast the output changes as the input changes

A sensitivity

Nudge the input by a small Δx and the output moves by about $f'(x)\,\Delta x$

- The **sign** gives the direction:
  - **Positive**: when the input increases, the output increases
  - **Negative**: when the input increases, the output decreases
- The **size** gives the strength: the greater the derivative, the greater the **variation** of the output when the input varies
- In machine learning the third interpretation is the most relevant, as $E'_{\text{in}}(\theta)$ answers the question: **if I change this parameter a little, how much does the error change, and in which direction?**

The derivative is itself a function



- There is one slope of the tangent line at each point, so the derivative is **itself a function**
- For the running example, $E'_{\text{in}}(\theta) = 15\,\theta - 30$ (computed two slides ago):

θ	0	1	2	3	4
$E'_{\text{in}}(\theta)$	-30	-15	0	15	30

- Left of 2 the derivative is **negative** and the error is decreasing. Right of 2 it is **positive** and the error is increasing
- At 2 it is **zero**, and the curve is flat

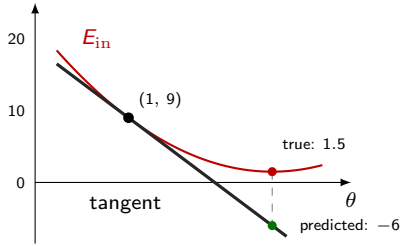
Zoom in, and every smooth curve is a line

- From the definition of derivative, for a small Δx the average rate of change is *almost* the derivative, and rearranging gives

Local linear approximation

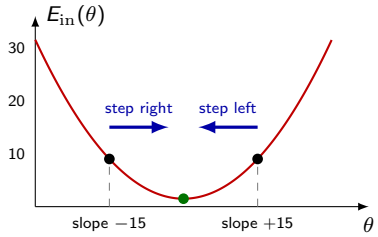
$$f(x + \Delta x) \;\approx\; f(x) + f'(x)\,\Delta x \qquad \text{for small } \Delta x$$

- In words: **new value** $\approx$ **old value** $+$ **slope** $\times$ **step**. Near x the function behaves like its tangent line
- Running example at $\theta = 1$, where $E_{\text{in}}(1) = 9$ and $E'_{\text{in}}(1) = -15$. A step of 0.1 predicts $E_{\text{in}}(1.1) \approx 9 - 1.5 = 7.5$. The true value is 7.575
- A step of 1 predicts $E_{\text{in}}(2) \approx 9 - 15 = -6$. The true value is 1.5. **The approximation is local**: excellent for small steps, useless for large ones



Which way is downhill?

- We start from a point x and want to go **down**, that is, decrease the function value. By the formula of the last slide, a small step Δx changes the function by about $f'(x)\,\Delta x$
- To make that change **negative**, the step must have the sign **opposite** to the derivative. Slope positive: step to the left. Slope negative: step to the right
- Conclusion: **step in the direction of** $-f'(x)$

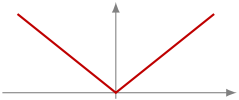


A first look at gradient descent

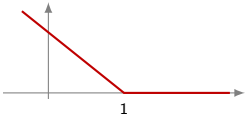
Replace x by $x - \alpha\, f'(x)$, for a small number $\alpha > 0$, and repeat.

- It uses only the derivative at the current point and decreases the value of the function as long as α is **small enough** and $f'(x) \neq 0$

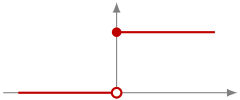
Where there is no derivative



$|x|$: a **corner** at 0



the hinge loss: a **corner** at 1



an indicator: a **jump**

- The derivative is a limit, and a limit may fail to exist:
  - **A corner**: at 0 the secants of $|x|$ have slope -1 on the left and $+1$ on the right, and there is no single tangent. The hinge loss $\max\{0, 1 - m\}$ has the same corner, at margin $m = 1$
  - **A jump**. The indicator is not even continuous. Differentiable functions are always continuous
- A corner is a mild problem, since the derivative exists everywhere else, while a jump is fatal. **This is why losses are designed to be continuous, and smooth wherever possible**

Rules of differentiation

How to compute derivatives

- Derivatives of functions given by a formula are computed from

A short table

The derivatives of a few basic functions, computed from the definition of derivative

A few rules

How to differentiate a function built from the basic functions

- If we know the derivatives of basic functions and how to compute the derivatives of their sums, products and compositions, then we know how to compute the derivative of a big class of functions
- With the table and the three rules we can differentiate every loss in this course, by hand when it is simple, with a computer when it is not

Table of derivatives

$f(x)$	$f'(x)$
c (a constant)	0
x	1
x^n	$n\,x^{n-1}$
e^x	e^x
$\log x$	$\frac{1}{x}$

- **Constant**: a function that never changes has rate of change 0
- **Power**: bring the power down and reduce it by one. It holds for *any* power: $\frac{1}{x} = x^{-1}$ has derivative $(-1)x^{-2} = -\frac{1}{x^2}$, and $\sqrt{x} = x^{1/2}$ has derivative $(1/2)x^{-1/2} = \frac{1}{2\sqrt{x}}$
- **Exponential**: e^x is its own derivative. The larger it is, the faster it grows
- **Logarithm**: the slope $1/x$ shrinks as x grows. The logarithm keeps increasing, ever more slowly
- Each derivative follows from the definition, exactly as x^2 did, but we omit the proof

Derivatives of sums and multiplication by constants

Proposition (Sum and constant multiple)

If f and g are differentiable and c is a constant, then $(f+g)' = f' + g'$ *and* $(c\,f)' = c\,f'$.

- Running example:

$$E_{\text{in}}(\theta) = 7.5\,\theta^2 - 30\,\theta + 31.5 \qquad \Longrightarrow \qquad E'_{\text{in}}(\theta) = 7.5 \times 2\theta - 30 \times 1 + 0 = 15\,\theta - 30$$

- An in-sample error is an average of one loss per observation, $E_{\text{in}}(\theta) = \frac{1}{N}\sum_{i=1}^{N} \ell_i(\theta)$ with $\ell_i(\theta) = (y_i - \theta x_i)^2$
- By the rule, each observation contributes its own derivative:

The derivative of an average is the average of the derivatives

$$E'_{\text{in}}(\theta) \;=\; \frac{1}{N}\sum_{i=1}^{N} \ell'_i(\theta)$$

The product rule

Proposition (Product rule)

If f and g are differentiable, then $(fg)'(x) = f'(x)\,g(x) + f(x)\,g'(x)$.

- It is **not** the product of the derivatives. For $x^2 = x \times x$ the rule gives $1 \times x + x \times 1 = 2x$, as in the table, while the product of the derivatives would give 1

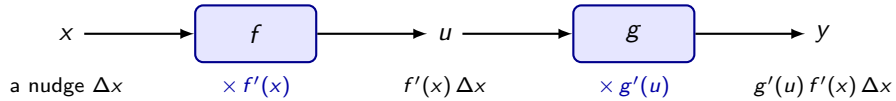
function	f	g	**derivative,** $f'g + f\,g'$
$x^2 e^x$	x^2	e^x	$2x\,e^x + x^2 e^x$
$x \log x$	x	$\log x$	$1 \times \log x + x \times \frac{1}{x} = \log x + 1$
$p\,(1-p)$	p	$1-p$	$1 \times (1-p) + p \times (-1) = 1 - 2p$

The chain rule

Theorem (Chain rule)

If $u = f(x)$ and $y = g(u)$, with f and g differentiable, then the composition $y = g\big(f(x)\big)$ has derivative

$$(g \circ f)'(x) \;=\; g'\big(f(x)\big)\, f'(x), \qquad \text{or, in the other notation,} \qquad \frac{dy}{dx} \;=\; \frac{dy}{du}\,\frac{du}{dx}.$$



- **Rates multiply**. If u changes 3 times as fast as x, and y changes 2 times as fast as u, then y changes $2 \times 3 = 6$ times as fast as x
- **Beware**: g' is evaluated at $u = f(x)$, the value coming *out* of the first function, not at x
- In machine learning this is the most important rule

The chain rule, in three steps

(1) Name the inside function u

(2) Differentiate the outside with respect to u

(3) Multiply by the derivative of the inside with respect to the input variable

function	**inside u**	**outside**	**derivative**
$(x^3-1)^{100}$	x^3-1	u^{100}	$100\,(x^3-1)^{99}3x^2$
e^{-x^2}	$-x^2$	e^u	$e^{-x^2}(-2x)$
$\log(1-\theta)$	$1-\theta$	$\log u$	$\frac{1}{1-\theta}(-1)$
e^{-z}	$-z$	e^u	$e^{-z}(-1)$

- Without the chain rule, the first line would need expanding a polynomial of degree 300
- **Write down the details of the computations in the table for you to practise**

The chain rule on a loss

- One observation (x, y), the model $h_\theta(x) = \theta x$ and the squared loss give $\ell(\theta) = (y - \theta x)^2$. Here x and y are **fixed numbers** and the variable is θ
- **Inside:** the residual $u = y - \theta x$, with derivative $-x$
- **Outside:** u^2, with derivative $2u$
- **Multiply:**

$$\ell'(\theta) \;=\; 2\,(y - \theta x)(-x) \;=\; -2\,x\,(y - \theta x)$$

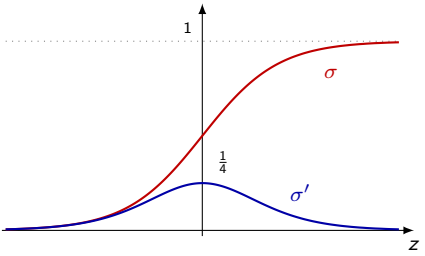
Residual times input

The derivative of the squared loss at one observation is $-2 \times$ **input** $\times$ **residual**

- Averaging over the sample: $E'_{\text{in}}(\theta) = -\dfrac{2}{N}\displaystyle\sum_{i=1}^{N} x_i\,(y_i - \theta x_i)$
- Running example: $-\frac{2}{4}\big[(2-\theta) + 2(5-2\theta) + 3(4-3\theta) + 4(9-4\theta)\big] = 15\,\theta - 30$. The same answer as before, by a different path

The logistic function and its derivative

- Write σ for the logistic curve of Lecture 4, as a function of the score z: $\sigma(z) = \dfrac{1}{1+e^{-z}}$
- A composition: inside $u = 1 + e^{-z}$, outside $1/u$. The chain rule gives $\sigma'(z) = \dfrac{e^{-z}}{(1+e^{-z})^2}$, which reduces to (**why?**)



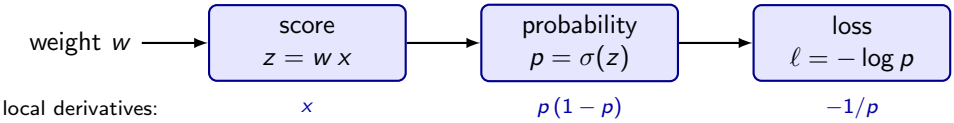
The derivative of the logistic function

$$\sigma'(z) \;=\; \sigma(z)\,\big(1-\sigma(z)\big)$$

- The slope is largest at $z = 0$, where $\sigma(0) = \frac{1}{2}$ and $\sigma'(0) = \frac{1}{4}$. Far from 0 the curve is flat and the derivative nearly 0

Long chains and automatic differentiation

- A model can be a **long** composition, and the chain rule extends link by link: the derivative of the whole pipeline is the **product of the local derivatives**



- Multiply the three: $\dfrac{d\ell}{dw} = -\dfrac{1}{p}\,p\,(1-p)\,x = -(1-p)\,x$. Each box needed only **its own** derivative and the value of the input
- That is mechanical and there is a general algorithm, called **automatic differentiation**, that is the engine inside popular ML libraries such as PyTorch, JAX and TensorFlow
- On a neural network, with the links multiplied from the loss backwards, it has a famous name: **backpropagation**

The derivative rules

$f(x)$	$f'(x)$
c	0
x^n	$n\,x^{n-1}$
e^x	e^x
$\log x$	$1/x$

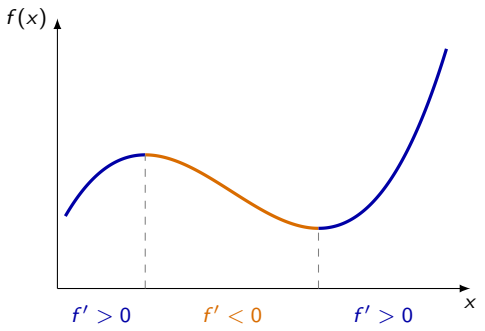
built as	**derivative**
a sum, $f+g$	$f'+g'$
a constant multiple, $c\,f$	$c\,f'$
a product, $f\,g$	$f'g+f\,g'$
a composition, $g\big(f(x)\big)$	$g'\big(f(x)\big)\,f'(x)$

- A quotient needs no rule of its own: write f/g as $f \times g^{-1}$ and use the product and chain rules
- Three derivatives of this section that we will use later:

$$(y-\theta x)^2 \;\longrightarrow\; -2\,x\,(y-\theta x) \qquad \log\theta \;\longrightarrow\; \frac{1}{\theta}, \qquad \sigma(z) \;\longrightarrow\; \sigma(z)\big(1-\sigma(z)\big)$$

Maxima and minima

Increasing and decreasing

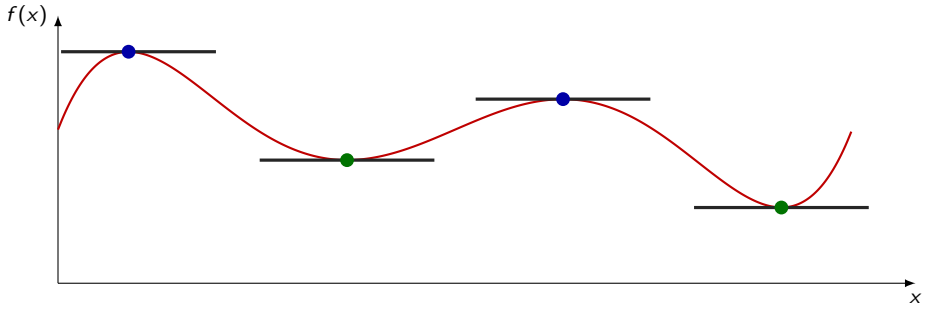


Proposition (Increasing and decreasing)

If $f'(x) > 0$ for every x in an interval, then f is **increasing** *on that interval. If $f'(x) < 0$, then f is* **decreasing**.

- Running example: $E'_{\text{in}}(\theta) = 15\,\theta - 30$ is negative for $\theta < 2$ and positive for $\theta > 2$. The error decreases until $\theta = 2$ and increases from there on
- The points where the derivative is zero, such as $\theta = 2$, are called **critical points**

Critical points



- **Question 1**: what do the four points have in common?
- **Question 2**: And what do the green points, and the blue points, have in common?

Global and local minimum

Definition (Minimum)

f has a **global minimum** at c if $f(c) \leq f(x)$ for **every** x. It has a **local minimum** at c if $f(c) \leq f(x)$ for every x **near** c.

- Every global minimum is a local one. The last picture shows that the converse fails
- The **minimiser** c is *where* the function is smallest and the **minimum value** $f(c)$ is *how small* it is. In symbols, $c = \arg\min_x f(x)$ and $f(c) = \min_x f(x)$
- In learning we are after the minimiser: $\hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} E_{\text{in}}(\boldsymbol{\theta})$ is the fitted model, and the minimum value is its in-sample error
- What we want is the **global** minimiser. What calculus can see, as we are about to find out, is only **local**
- The same idea applies to **global and local maximums**

At local minimums the derivative is zero

Theorem (Fermat)

If f has a local minimum or a local maximum at c, and f is differentiable at c, then $f'(c) = 0$.

- **Why**:
  - if $f'(c)$ were positive, a small step to the left would decrease the function and
  - if $f'(c)$ were negative, a small step to the right would decrease the function

  Either way, c would not be a local minimum

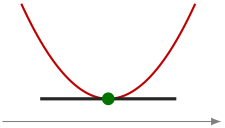
How to find a minimiser

To find the minimisers of a differentiable function, first **solve the equation** $f'(x) = 0$ and then compute the value of f at all the solutions to find the global minimum

- An optimisation problem has become an **equation**. Sometimes it can be solved with pen and paper, but more often than not it needs an algorithm

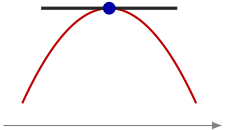
But zero derivative does not always mean local minimum

- At a critical point three things can happen:



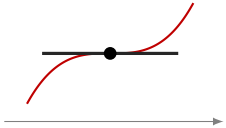
a local minimum

f' goes from $-$ to $+$



a local maximum

f' goes from $+$ to $-$



neither, as for x^3 at 0

f' keeps its sign

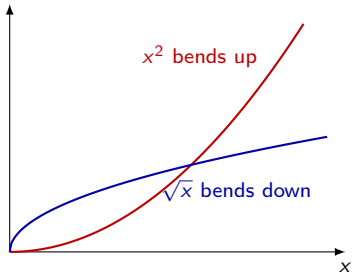
- The sign of f' on each side tells the three apart and provides a test for a local minimum: the **first derivative test**
- And a local minimum need not be the global one. So $f'(x) = 0$ only produces **candidates**

The second derivative

Definition (Second derivative)

The **second derivative** f'' is the derivative of the derivative f'. It is the rate of change of the **slope**.

- $f''(x) > 0$: the slope is increasing and the curve **bends up**. $f''(x) < 0$: the slope is decreasing and the curve **bends down**
- Both curves in the picture are increasing. x^2 has $f'' = 2 > 0$ and bends up; $\sqrt{x}$ has $f'' = -\frac{1}{4}x^{-3/2} < 0$ and bends down
- Running example: $E'_{\text{in}}(\theta) = 15\,\theta - 30$, so $E''_{\text{in}}(\theta) = 15$
- At a critical point the slope is zero, but was it increasing or decreasing as it approached the critical point?

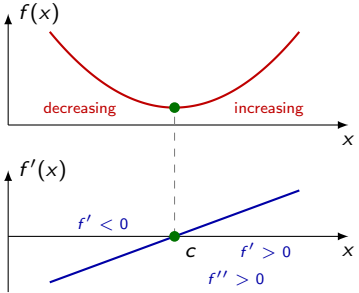


The second derivative test

Proposition (Second derivative test)

If $f'(c) = 0$ and $f''(c) > 0$, then c is a local minimiser. If $f'(c) = 0$ and $f''(c) < 0$, then c is a local maximiser.

- **Why**: at a critical point the slope is zero, and $f''(c)$ says how the slope varies as it passes through zero
- $f''(c) > 0$: the slope is **increasing**, negative just before c and positive just after. So f decreases and then increases: a local minimum
- $f''(c) < 0$: the slope is **decreasing**, so f increases and then decreases: a local maximum



- Running example: $E''_{\text{in}}(2) = 15 > 0$, so $\theta = 2$ is a local minimum. If $f''(c) = 0$ the test says nothing: compare x^3 and x^4 at 0

Example: maximum likelihood

- Lecture 4: 7 heads in 10 tosses, with likelihood $\mathcal{L}(\theta) = 120\,\theta^7(1-\theta)^3$. We want the θ that maximises it
- Since log is increasing, $\mathcal{L}$ and $\log \mathcal{L}$ are largest at the **same** θ, which is where $-\log \mathcal{L}$ is **smallest**. So we **minimise** minus the log-likelihood, and the logarithm turns the product into a sum:

$$-\mathrm{LL}(\theta) \;=\; -\log 120 - 7\log\theta - 3\log(1-\theta)$$

- Differentiate term by term, with the table and the chain rule for the last term:

$$(-\mathrm{LL})'(\theta) \;=\; -\frac{7}{\theta} + \frac{3}{1-\theta}$$

The answer of Lecture 4, now derived

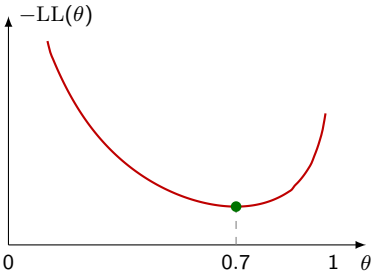
Set it to zero: $3\,\theta = 7\,(1-\theta)$, that is, $10\,\theta = 7$ and $\hat{\theta} = \dfrac{7}{10}$

Example: maximum likelihood

- Nothing was special about 7 and 10. With k successes in N independent trials, $-\mathrm{LL}(\theta) = \text{constant} - k\log\theta - (N-k)\log(1-\theta)$ and

$$(-\mathrm{LL})'(\theta) \;=\; -\frac{k}{\theta} + \frac{N-k}{1-\theta} \;=\; 0 \qquad \Longleftrightarrow \qquad \hat{\theta} = \frac{k}{N}$$

- The **observed proportion** of successes: the formula Lecture 4 stated without derivation
- Is it a minimum? $(-\mathrm{LL})''(\theta) = \frac{k}{\theta^2} + \frac{N-k}{(1-\theta)^2} > 0$, so the slope is increasing as it passes through zero: **a local minimum** of $-\mathrm{LL}$, by the second derivative test



The coin: $k = 7$, $N = 10$

- The same three steps (take logarithms, differentiate, set to zero) give the sample mean and the sample variance for the Normal family, the **maximum likelihood** estimators of Lecture 4

Example: least squares with one parameter

- For the line through the origin, $h_\theta(x) = \theta x$: differentiate the in-sample error, set the derivative to zero, solve for θ, and check the second derivative

Least-squares estimate

$$E_{\text{in}}(\theta) = \frac{1}{N}\sum_{i=1}^{N}\left(y_i - \theta x_i\right)^2$$

$$E'_{\text{in}}(\theta) = -\frac{2}{N}\sum_{i=1}^{N} x_i\left(y_i - \theta x_i\right) = 0 \iff \sum_{i=1}^{N} x_i y_i = \theta \sum_{i=1}^{N} x_i^2 \iff \hat{\theta} = \frac{\sum_i x_i y_i}{\sum_i x_i^2}$$

$$E''_{\text{in}}(\theta) = \frac{2}{N}\sum_{i=1}^{N} x_i^2 > 0 \implies \hat{\theta} \text{ is a } \textbf{local minimum}, \text{ by the second derivative test}$$

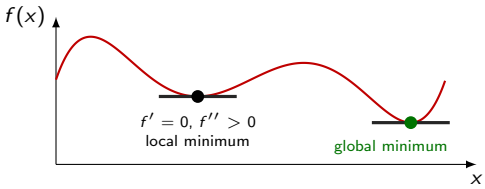
- Running example: $\hat{\theta} = \dfrac{2+10+12+36}{1+4+9+16} = \dfrac{60}{30} = 2$ and $E''_{\text{in}}(\theta) = \dfrac{2}{4} \times 30 = 15 > 0$

Two problems supposedly solved. But are these local minimums really the **global** *minimums?*

Convexity

How do we know a local minimum is the global minimum?

- **What we have**: $f'(c) = 0$ and $f''(c) > 0$ give a local minimum. Both are tests **at one point**
- **What we want**: the smallest value over **all** x
- In the picture two points pass both tests. Only one is the global minimum



	$f'(c) = 0$	$f''(c) > 0$	$f'' > 0$ **at every point**
The coin: $-\mathrm{LL}$ at $\hat{\theta} = 7/10$	yes	yes	yes: $\frac{7}{\theta^2} + \frac{3}{(1-\theta)^2}$
The line: E_{in} at $\hat{\theta} = 2$	yes	yes	yes: 15
The curve of the picture, at the black point	yes	yes	**no**

What we are looking for

A property of the **whole function** which guarantees in advance that **zero derivative means global minimum**: **convexity**.

Convex and concave functions

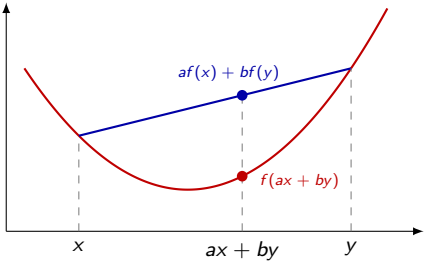
Definition (Convex function)

A function f is **convex** if, for any two points x and y and any weights $a, b \geq 0$ with $a + b = 1$,

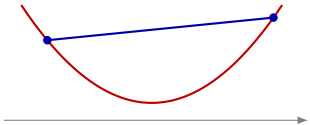
$$f(ax + by) \leq af(x) + bf(y).$$

It is **concave** if $-f$ is convex.

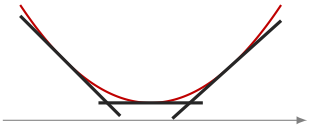
- $ax + by$ is a weighted average of x and y: a point **between** them. The left side is the function at that point. The right side is the same average of the two heights, which is the height of the **chord** joining the two points of the graph
- In one sentence: **every chord lies above the graph**



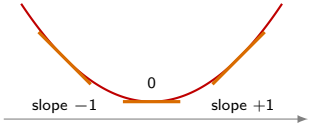
Three ways to see convexity



every **chord** lies above the graph



the graph lies above every **tangent**



the slope never decreases: $f'' \geq 0$

- For well-behaved functions the three statements are **equivalent**: they describe the same shape
- The third is the practical test. Compute f'' and check its sign
- Notice what kind of statement this is. Not "at this point", but **every** chord, **every** tangent, **every** x. Convexity is a property of the whole function

A small catalogue

convex	**concave**	**neither**
x^2, since $f'' = 2$	$\log x$, since $f'' = -1/x^2$	x^3: $f'' = 6x$ changes sign
e^x, since $f'' = e^x$	$\sqrt{x}$	the logistic curve $\sigma(z)$
$-\log x$, since $f'' = 1/x^2$	$-x^2$	the binary loss (a staircase)
$\lvert x \rvert$ and the hinge $\max\{0, 1-m\}$		

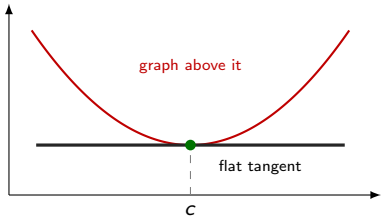
- A sum of convex functions is convex
- A positive multiple of a convex function is convex
- If f is convex, $f(a\theta + b)$ is convex in θ for any $a, b \in \mathbb{R}$
- So $E_{\text{in}}(\theta) = \frac{1}{N}\sum_i (y_i - \theta x_i)^2$ is convex: each term is the convex function u^2 of a line in θ, and the terms are added

Minimums of convex functions

Theorem

If f is convex and differentiable, and $f'(c) = 0$, then c is a **global** *minimiser of f.*

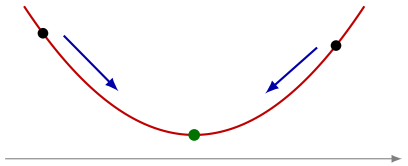
- **Why**: a convex function lies above all its tangents. The tangent at c is flat, at height $f(c)$. So $f(x) \geq f(c)$ for every x
- Our two examples: $-\mathrm{LL}$ and E_{in} have $f'' > 0$ at every θ, so they are convex. Hence $\hat{\theta} = 7/10$ and $\hat{\theta} = 2$ are **global** minimisers



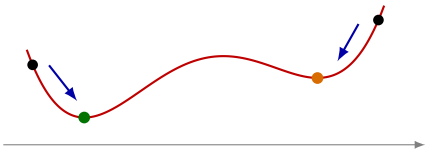
Convex: every local minimum is global

For a convex function, solving $f'(x) = 0$ gives **the answer**, not a list of candidates

Convex and non-convex, in a picture



convex: one valley



not convex: where we end depends on where we start

- On the left, any starting point and any downhill path lead to the same place. On the right, the start on the right is trapped in the orange valley: a **local** minimum that is not the global one
- **Convex** in this course: least squares, ridge regression, logistic regression, the SVM
- **Not convex**: K-means and neural networks

Several variables

Real models have more than one parameter

- Put the intercept back into the running example: $h_{\boldsymbol{\theta}}(x) = \theta_0 + \theta_1 x$. The error now depends on **two** numbers,

$$E_{\text{in}}(\theta_0, \theta_1) \;=\; \frac{1}{4}\sum_{i=1}^{4}\left(y_i - \theta_0 - \theta_1 x_i\right)^2 \;=\; \theta_0^2 + 5\,\theta_0\theta_1 + 7.5\,\theta_1^2 - 10\,\theta_0 - 30\,\theta_1 + 31.5$$

- One error for each *pair* of parameters:

(θ_0, θ_1)	$(0,0)$	$(1,1)$	$(2,1)$	$(0,2)$
$E_{\text{in}}(\theta_0, \theta_1)$	31.5	5	3	1.5

- In general a model has m parameters and the error is a function $f : \mathbb{R}^m \to \mathbb{R}$: the input is a **vector** $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_m)$, the output is still one number
- As with integrals in Lecture 2, **the idea behind derivatives is unchanged** when we increase the input dimension

Partial derivatives

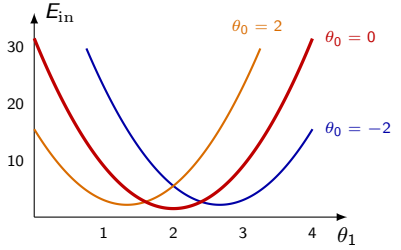
Definition (Partial derivative)

The **partial derivative** of f with respect to x_j, written $\frac{\partial f}{\partial x_j}$, is the ordinary derivative of f in the variable x_j, with all the other variables held **constant**.

- Treat the frozen variables as numbers and use the rules we know
- For the running example:

$$\frac{\partial E_{\text{in}}}{\partial \theta_0} = 2\,\theta_0 + 5\,\theta_1 - 10, \qquad \frac{\partial E_{\text{in}}}{\partial \theta_1} = 5\,\theta_0 + 15\,\theta_1 - 30$$

- **In the picture**: if we fix θ_0, the error is a parabola in θ_1



The gradient

Definition (Gradient)

The **gradient** of $f : \mathbb{R}^m \to \mathbb{R}$ at $\mathbf{x}$ is the vector of its partial derivatives,

$$\nabla f(\mathbf{x}) \;=\; \Big(\frac{\partial f}{\partial x_1}(\mathbf{x}),\; \ldots,\; \frac{\partial f}{\partial x_m}(\mathbf{x})\Big)^T \;\in\; \mathbb{R}^m.$$

- One entry per variable: the sensitivity of f to each of its inputs, collected in a **vector**. With one variable, the gradient is just $f'(x)$
- Running example: $\nabla E_{\text{in}}(\theta_0, \theta_1) = \big(2\,\theta_0 + 5\,\theta_1 - 10,\;\; 5\,\theta_0 + 15\,\theta_1 - 30\big)^T$
- At $(2, 2)$ the gradient is $(4, 10)^T$: there the error is more than twice as sensitive to the slope as to the intercept. At $(0, 2)$ it is $(0, 0)^T$

Gradients and linear approximation

Linear approximation in several variables

$$f(\mathbf{x}+\Delta\mathbf{x}) \;\approx\; f(\mathbf{x}) + \nabla f(\mathbf{x})^T \Delta\mathbf{x} \qquad \text{for a small step } \Delta\mathbf{x}$$

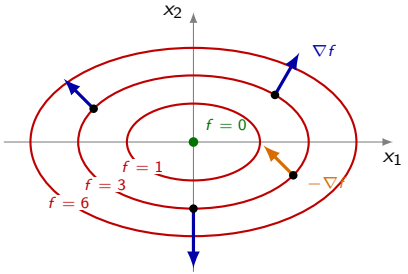
- It is $f(x+\Delta x) \approx f(x) + f'(x)\,\Delta x$ with the product replaced by an **inner product**:

$$\nabla f(\mathbf{x})^T \Delta\mathbf{x} \;=\; \frac{\partial f}{\partial x_1}\Delta x_1 + \cdots + \frac{\partial f}{\partial x_m}\Delta x_m$$

- Each variable contributes its own sensitivity times its own step, and the contributions add
- Running example at $(2,2)$, where $E_{\text{in}} = 5.5$ and $\nabla E_{\text{in}} = (4,10)^T$. The step $\Delta\boldsymbol{\theta} = (-0.1,-0.1)^T$ predicts a change of $4\times(-0.1) + 10\times(-0.1) = -1.4$, that is $E_{\text{in}} \approx 4.1$. The true value is 4.235

The gradient points uphill

- A step of small length s along a unit vector $\mathbf{u}$ gives $f(\mathbf{x} + s\,\mathbf{u}) \approx f(\mathbf{x}) + s\,\nabla f(\mathbf{x})^T \mathbf{u}$. **Which direction makes f increase fastest?**
- Cauchy–Schwarz answers: $\nabla f(\mathbf{x})^T \mathbf{u} \leq \|\nabla f(\mathbf{x})\|$, with equality exactly when $\mathbf{u} = \|\nabla f\|^{-1} \nabla f(\mathbf{x})$
- Each curve of the picture is a **level curve**, the points where f takes the same value. Along it f does not change, so $\nabla f(\mathbf{x})^T \mathbf{u} = 0$: the gradient is **perpendicular to the level curves**



Level curves of $f(x_1, x_2) = x_1^2 + 3x_2^2$

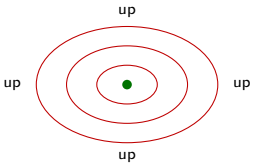
Steepest ascent, steepest descent

$\nabla f(\mathbf{x})$ is the direction of **steepest ascent**, and its length $\|\nabla f(\mathbf{x})\|$ is the slope in that direction. The opposite direction, $-\nabla f(\mathbf{x})$, is the direction of **steepest descent**.

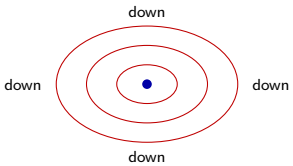
Critical points, local minimums and maximums

Theorem (Fermat, in several variables)

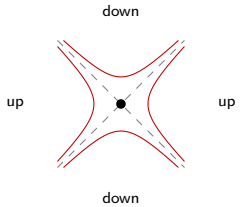
If f has a local minimum or a local maximum at $\mathbf{x}$, and f is differentiable there, then $\nabla f(\mathbf{x}) = \mathbf{0}$.



a **minimum**



a **maximum**



a **saddle**

- A point with $\nabla f(\mathbf{x}) = \mathbf{0}$ is a **critical point**
- The **saddle** is new, with the function increasing in some directions and decreasing in others

Example: the line with an intercept

- Set both partial derivatives of $E_{\text{in}}(\theta_0, \theta_1) = \frac{1}{N}\sum_i (y_i - \theta_0 - \theta_1 x_i)^2$ to zero: two **linear** equations in two unknowns. Solve the first for θ_0 and substitute it in the second

The least-squares line

$$\frac{\partial E_{\text{in}}}{\partial \theta_0} = -\frac{2}{N}\sum_i \big(y_i - \theta_0 - \theta_1 x_i\big) = 0 \quad \Longleftrightarrow \quad \theta_0 = \bar{y} - \theta_1 \bar{x}$$

$$\frac{\partial E_{\text{in}}}{\partial \theta_1} = -\frac{2}{N}\sum_i x_i\big(y_i - \theta_0 - \theta_1 x_i\big) = 0 \quad \Longrightarrow \quad 0 = \sum_i x_i\big[(y_i - \bar{y}) - \theta_1(x_i - \bar{x})\big]$$

$$\hat{\theta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}, \qquad \hat{\theta}_0 = \bar{y} - \hat{\theta}_1 \bar{x}$$

- The slope is the one-parameter formula applied to the **centred** data $x_i - \bar{x}$ and $y_i - \bar{y}$
- Running example: $\bar{x} = 2.5$, $\bar{y} = 5$, so $\hat{\theta}_1 = 10/5 = 2$ and $\hat{\theta}_0 = 5 - 2 \times 2.5 = 0$. On this data the best line happens to pass through the origin

The only critical point. Is it a minimum, a maximum or a saddle?

Gradients in matrix notation

function of $\mathbf{x} \in \mathbb{R}^m$	gradient	in one variable
$\mathbf{a}^T\mathbf{x}$	$\mathbf{a}$	$(a\,x)' = a$
$\|\mathbf{x}\|^2 = \mathbf{x}^T\mathbf{x}$	$2\,\mathbf{x}$	$(x^2)' = 2x$
$\mathbf{x}^T A\,\mathbf{x}$, with A symmetric	$2A\,\mathbf{x}$	$(a\,x^2)' = 2a\,x$

- **Check the first**: $\mathbf{a}^T\mathbf{x} = a_1x_1 + \cdots + a_mx_m$ has partial derivative a_j in x_j. Stack them and you get $\mathbf{a}$
- **Check the third** for $m = 2$: $\mathbf{x}^T A\,\mathbf{x} = a_{11}x_1^2 + 2a_{12}x_1x_2 + a_{22}x_2^2$ has partial derivatives $2a_{11}x_1 + 2a_{12}x_2$ and $2a_{12}x_1 + 2a_{22}x_2$, the two entries of $2A\,\mathbf{x}$
- **The sum and multiplication by constant rules hold as in one dimension**

The gradient of the residual sum of squares

- The responses are a vector $\mathbf{Y} \in \mathbb{R}^N$, and the data matrix is $X \in \mathbb{R}^{N \times m}$, with one **row** per observation. The predictions are $X\boldsymbol{\theta}$ and the residual sum of squares is $\|\mathbf{Y} - X\boldsymbol{\theta}\|^2$
- With the derivative table of the previous slide in mind,

$$\mathrm{RSS}(\boldsymbol{\theta}) = \|\mathbf{Y} - X\boldsymbol{\theta}\|^2 = \underbrace{\boldsymbol{\theta}^T X^T X \boldsymbol{\theta}}_{\text{third row, } A = X^T X} - \underbrace{2\mathbf{Y}^T X \boldsymbol{\theta}}_{\text{first row, } \mathbf{a} = X^T \mathbf{Y}} + \underbrace{\mathbf{Y}^T \mathbf{Y}}_{\text{constant}}$$

The gradient of the RSS

$$\nabla \mathrm{RSS}(\boldsymbol{\theta}) = 2X^T X \boldsymbol{\theta} - 2X^T \mathbf{Y} = -2X^T(\mathbf{Y} - X\boldsymbol{\theta})$$

- $\mathbf{Y} - X\boldsymbol{\theta}$ is the vector of residuals, and X^T multiplies it by the inputs. It is **"residual times input"** for every parameter and every observation at once
- Running example, where the rows of X are $(1, x_i)$: $X^T X = \begin{pmatrix} 4 & 10 \\ 10 & 30 \end{pmatrix}$ and $X^T \mathbf{Y} = \begin{pmatrix} 20 \\ 60 \end{pmatrix}$. Dividing by $N = 4$ recovers the gradient of the last slides

The Hessian matrix

Definition (Hessian)

The **Hessian** of f at $\mathbf{x}$ is the $m \times m$ matrix $\nabla^2 f(\mathbf{x})$ of its second partial derivatives: the entry in row j and column k is $\frac{\partial^2 f}{\partial x_j \, \partial x_k}(\mathbf{x})$. For well-behaved f it is **symmetric**.

- Row j records how the j-th partial derivative changes as each variable moves: the Hessian is the gradient of the gradient
- Two examples from the last slides: $\mathbf{x}^T A \mathbf{x}$ has Hessian $2A$, and the RSS has Hessian $2X^T X$
- Running example: $\nabla^2 E_{\text{in}} = \begin{pmatrix} 2 & 5 \\ 5 & 15 \end{pmatrix}$, the same matrix at every point
- **Curvature in a direction**. Along a unit vector $\mathbf{u}$ the surface bends like the quadratic form $\mathbf{u}^T \nabla^2 f(\mathbf{x}) \, \mathbf{u}$. Along the slope axis, $\mathbf{u} = (0, 1)^T$, this gives 15: the E''_{in} of the one-parameter example

Convex functions of several variables

- The definition does not change: $f(a\,\mathbf{x} + b\,\mathbf{y}) \leq a\,f(\mathbf{x}) + b\,f(\mathbf{y})$ for any two points of $\mathbb{R}^m$ and any weights $a, b \geq 0$ with $a + b = 1$

Theorem (Convexity and the Hessian)

A well-behaved function f is convex if and only if its Hessian is **positive semidefinite** *at every point: $\mathbf{u}^T \nabla^2 f(\mathbf{x})\,\mathbf{u} \geq 0$ for every $\mathbf{x}$ and every direction $\mathbf{u}$. And if f is convex and $\nabla f(\mathbf{x}) = \mathbf{0}$, then $\mathbf{x}$ is a* **global** *minimiser.*

- Recall that this is the same as saying that all eigenvalues of the Hessian are non-negative
- Running example: the eigenvalues of $\begin{pmatrix} 2 & 5 \\ 5 & 15 \end{pmatrix}$ are about 16.7 and 0.3. Both are positive, so E_{in} is convex

Least squares is convex

- The Hessian of the RSS is $2X^TX$. Take any vector $\mathbf{v}$:

$$\mathbf{v}^T(2X^TX)\mathbf{v} \;=\; 2\,(X\mathbf{v})^T(X\mathbf{v}) \;=\; 2\,\|X\mathbf{v}\|^2 \;\geq\; 0$$

Least squares is a convex problem

Whatever the data and whatever the basis functions, the RSS is convex in $\boldsymbol{\theta}$. Every critical point is a global minimiser: solving $\nabla \mathrm{RSS}(\boldsymbol{\theta}) = \mathbf{0}$ **is** solving least squares.

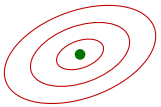
- Running example: $\frac{2}{N}X^TX = \begin{pmatrix} 2 & 5 \\ 5 & 15 \end{pmatrix}$, the Hessian of E_{in}. So $(0, 2)$, the only critical point, is a global minimiser
- Convexity says that a critical point is a global minimiser. It does not say that there is **only one**

When is the least-squares minimiser unique?

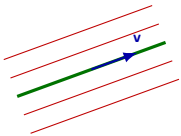
- Let $\hat{\boldsymbol{\theta}}$ be a critical point, $X^T(\mathbf{Y} - X\hat{\boldsymbol{\theta}}) = \mathbf{0}$, and move away from it by any vector $\mathbf{v}$:

$$\mathrm{RSS}(\hat{\boldsymbol{\theta}} + \mathbf{v}) \;=\; \|(\mathbf{Y} - X\hat{\boldsymbol{\theta}}) - X\mathbf{v}\|^2 \;=\; \mathrm{RSS}(\hat{\boldsymbol{\theta}}) \;-\; \underbrace{2\,\mathbf{v}^T X^T(\mathbf{Y} - X\hat{\boldsymbol{\theta}})}_{=\,0} \;+\; \|X\mathbf{v}\|^2$$

- Moving by $\mathbf{v}$ changes the predictions by $X\mathbf{v}$ and costs exactly $\|X\mathbf{v}\|^2$. So $\hat{\boldsymbol{\theta}} + \mathbf{v}$ is as good as $\hat{\boldsymbol{\theta}}$ only if $X\mathbf{v} = \mathbf{0}$: **the predictions do not change at all**



Columns of X linearly independent (full column rank). Then $X\mathbf{v} = \mathbf{0}$ only for $\mathbf{v} = \mathbf{0}$, and every other $\boldsymbol{\theta}$ is strictly worse: **one minimiser**



Columns linearly dependent. Some $\mathbf{v} \neq \mathbf{0}$ has $X\mathbf{v} = \mathbf{0}$, and $\hat{\boldsymbol{\theta}} + c\,\mathbf{v}$ is as good as $\hat{\boldsymbol{\theta}}$ for every c: **a line of minimisers**, which the data cannot tell apart

Level curves of the RSS in the plane of the parameters, with the minimisers in green

- Running example: the columns $(1,1,1,1)^T$ and $(1,2,3,4)^T$ of X are independent, so $(0,2)$ is the only minimiser. Had the four x_i been equal, every line through the point $(\bar{x}, \bar{y})$ would fit equally well

From one variable to many

	one variable	**several variables**
sensitivity	the derivative $f'(x)$, a number	the gradient $\nabla f(\mathbf{x})$, a vector
a small step	$f(x) + f'(x)\,\Delta x$	$f(\mathbf{x}) + \nabla f(\mathbf{x})^T \Delta \mathbf{x}$
downhill	the direction of $-f'(x)$	the direction of $-\nabla f(\mathbf{x})$
a critical point	$f'(x) = 0$	$\nabla f(\mathbf{x}) = \mathbf{0}$
bending	the second derivative $f''(x)$	the Hessian $\nabla^2 f(\mathbf{x})$, a matrix
convex	$f''(x) \geq 0$ everywhere	Hessian positive semidefinite everywhere
no traps	convex and $f'(x) = 0$: global minimiser	convex and $\nabla f(\mathbf{x}) = \mathbf{0}$: global minimiser

Lecture highlights

- **Functions and limits**: function, limit, continuous function
- **The derivative**: average rate of change, derivative, linear approximation
- **Rules of differentiation**: table of derivatives; sum, constant multiple, product and chain rules
- **Maxima and minima**: local and global minimum, critical point, first and second derivative tests
- **Convexity**: convex and concave functions, the criterion $f'' \geq 0$; a critical point of a convex function is a global minimiser
- **Several variables**: partial derivative, gradient, direction of steepest ascent, critical point, Hessian; a function is convex when its Hessian is positive semidefinite
- **Applications**: maximum likelihood estimator of a proportion; least-squares estimators; gradient, convexity and uniqueness of the minimiser of the residual sum of squares

Next: *Optimisation.*